UPPSALA
UNIVERSITET

Database Design Project

Course Name: Database Design
Course Code: 1DL301

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Git Repository: https://github.com/berof3/Database-Design

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1 ER Diagram

We are presenting the final version of our ER diagram.

The final version comprehends all the elements necessary to meet the base requirements + optional elements to meet one additional challenge.
The additional challenge is “Add support for order history (how and when the order status changed)” and the elements added to complete it are the weak entity “order_status_hist” and its identifying relationship “has” with entity “Order”

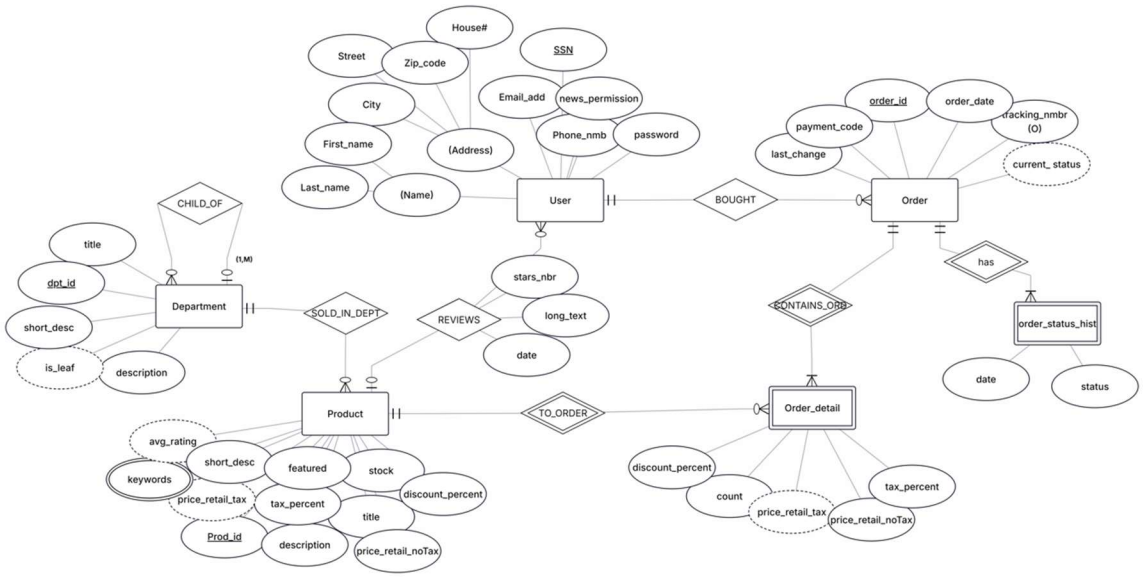


Figure 1 – ER Diagram (final version)

1.1 Department page and structure

CHILD_OF recursive relation has been added to Department entity. This will enable us to manage the hierarchical structure.
The same relationship will be used to define which departments are “leaf departments”, and the info will be stored in a derived attribute “is_leaf”.

Breadcrumbs will be obtained using the recursive relationship defined above.

1.2 Product

Breadcrumbs are managed using the breadcrumbs of the department to which the product belongs.

Retail price with tax will be a derived attribute (from price_retail_noTax and tax_percent).

Keywords will be managed as a multivalue attribute.

Average rating will be a derived attribute, calculated by selecting from the dedicated table all the reviews linked to the product.

1.3 Homepage

No need to add something in particular to manage them. Department entity is enough.

1.4 Users

User email will need to be unique. It is not a key, but a constraint will be put in place to check that there are no different profiles with the same email address.

1.5 Reviews

Reviews are managed by a relationship between the user and the product.

1.6 Orders

Tracking number is defined as optional attribute since its availability is not mandatory.

To keep track of prices we are using the “order_item” entity. Here, we will store the actual price taken from the product when the order was created.

1.7 Relationships

Relationships are defined in the diagram and their cardinality is self-explanatory.

2 Relational Schema

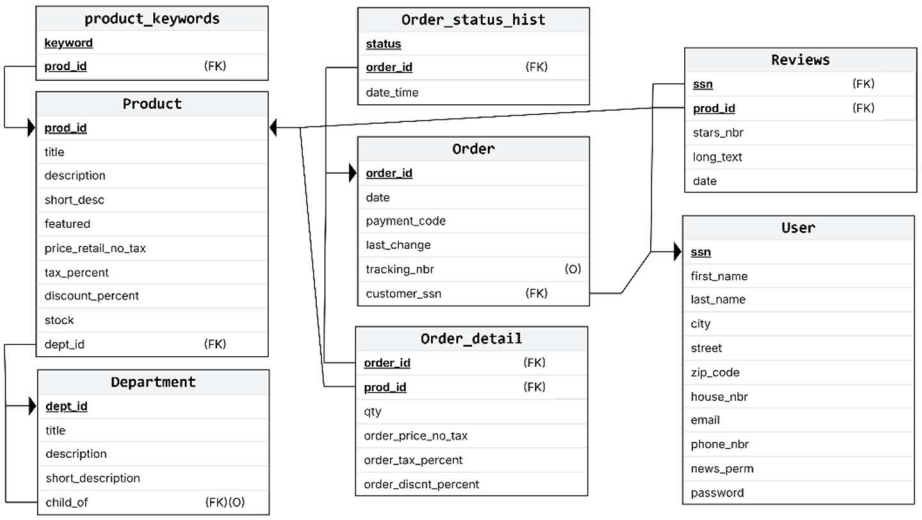


Figure 2 – Relational Schema

The image shows the final version of our relational schema. To obtain it, we started from the ER diagram and proceeded in the following order:

- First, we created the relations for the entities (not weak) defined in the diagram.

- For each entity, we then defined the attributes, inserting the "normal" ones first.

- For composite attributes, we inserted the atomic attributes of which they are composed.
  User.Name (ER diagram) → first_name/last_name (User, Relational schema)
  User.Address (ER diagram) → city/street/zip_code/house_nbr (User, Relational schema)

- Derived attributes were not inserted, as they will be calculated using the following logic:
  Department.is_leaf → It will be calculated by analysing the "child_of" column. If a department's "dept_id" does not appear in the column, it means that it is a leaf.

Product.avg_rating → Calculated by collecting values from the "Reviews" relation. It will be the average of the values in the "stars_nbr" column of all rows where Product.prod_id = Reviews.prod_id.
Product.price_retail_tax → From two attributes in Product, price_retail_no_tax + (price_retail_no_tax * tax_percent).
If discount_percent is not NULL it will be considered in the calculation.
Order_detail.price_retail_tax → From two attributes in Order_detail, price_retail_noTax + (price_retail_noTax * tax_percent).
If discount_percent is not NULL it will be considered in the calculation.
Order.current_status → The "order_id" will be used to collect all rows from the "Order_status_hist" relation. The status of the row with the most recent "date_time" will be retained.

- A relation was created for each multivalue attribute (just "Keywords")

- The "Reviews" relation was created to contain the link between users and comments regarding different products.

2.1 Pks and FKs

Relation	PK	Reason (if not clear)
User	SSN	--
Product	Prod_id	--
Department	Dept_id	--
Order	Order_id	--
Order_detail	Order_detial + Prod_id	It's a weak entity with two identifying relationships. The PK is made by the PK of both definying entities (Order and Product).
Order_status_hist	Order_id + status	"Order_id" is the PK of the definying entity, and status is the partial key that define the uniqueness of each row containing the same order_id.
product_keywords	Prod_id + keyword	--
Reviews	Ssn + prod_id	PKs of the two entities that are connected from the "reviewed" relationship. Since each user can only review a product once, the choice is self explanatory.

Relation	FK	Reason (if not clear)

User	--	--
Product	Dept_id	Define relationship "SOLD_IN_DEPT" with Department.
Department	Child_of	Define recursive relationship "CHILD_OF" with other Departments.
Order	Customer_ssn	Define relationship "BOUGHT" with Department.
Order_detail	Order_id/prod_id	Define relationship "TO_ORDER" with Product and "CONTAINS_ORDER" with Order.
Order_status_hist	Order_id	Define relationship "HAS" with Order.
product_keywords	Prod_id	Each product can have different keywords that help to search for it (or its suggestion based on other viewed products).
Reviews	Ssn / prod_id	Each review has two links, one with the user that wrote it and one with the product for which it was written.

3 Normalization

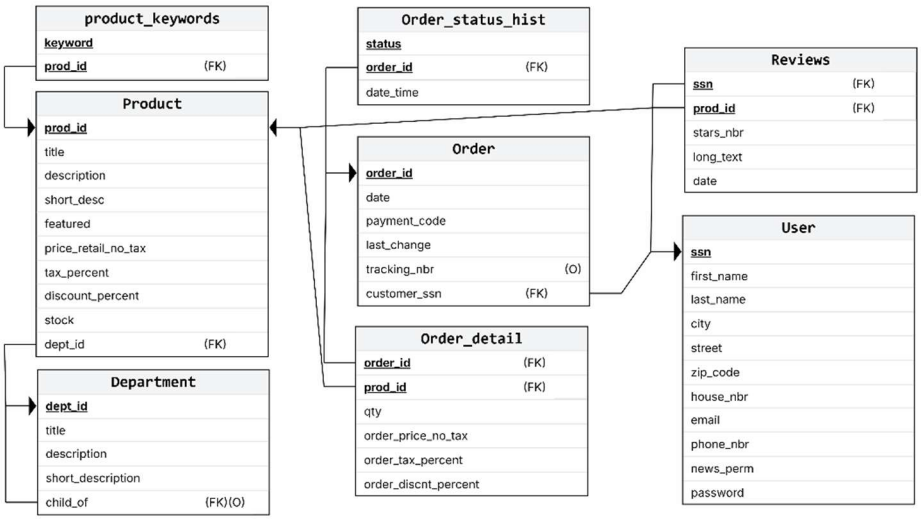


Figure 3 – 1NF/2NF/3NF

Let us recall the criteria by which we can define a relational schema in 1NF:

- There are no multivalued attributes.
- There are no composite attributes.
- Each value is atomic.
- Each record is uniquely identifiable by a PK.

in 2NF:

- is in 1NF
- every non-prime attribute is fully functionally dependent on the primary key (in other words, no non-key attribute depends only on part of the PK).

and in 3NF:

- is in 2NF
- there are no transitive functional dependencies between non-key attributes.

In our case, the three forms coincide.

In the transition from the ER diagram to the relational schema, all composite/multivalued/derived attributes were broken down or handled with dedicated relations.

Subsequently, no FDs were identified that required further normalisation of the schema to achieve 2NF and 3NF.

4 Tables Creation and Population

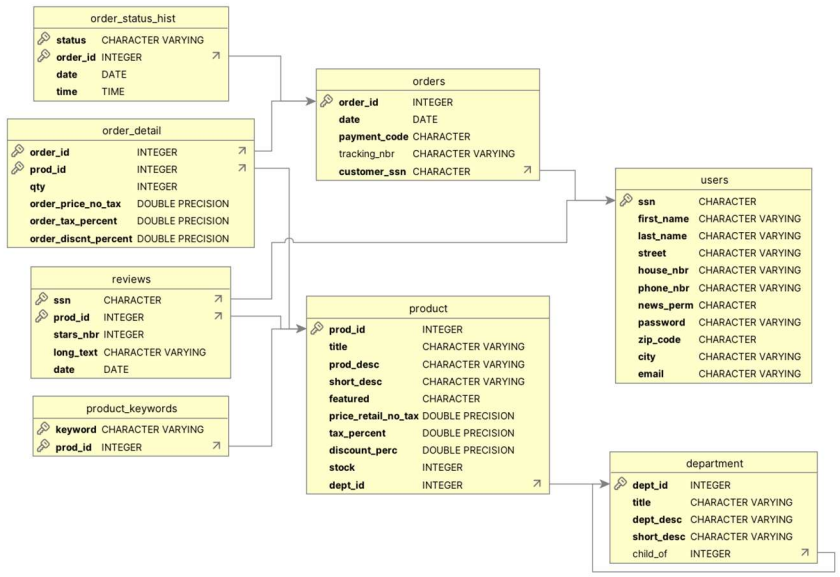
Once the relational schema is finished, a script for creating the tables is needed. The script follows the standard rules for creating tables; below can be found the code for creating all tables.

```
create table users
( ssn char(12) NOT NULL,
first_name varchar(30) NOT NULL,
last_name varchar(30) NOT NULL,
street varchar(30) NOT NULL,
house_nbr varchar(5) NOT NULL,
phone_nbr varchar(20) NOT NULL,
news_perm char(1) NOT NULL,
password varchar(50) NOT NULL,
zip_code char(5) NOT NULL,
city varchar(40) NOT NULL,
email varchar(40) NOT NULL,
primary key (ssn),
unique (email)
);

create table department
( dept_id int NOT NULL,
title varchar(30) NOT NULL,
dept_desc varchar(200) NOT NULL,
short_desc varchar(50) NOT NULL,
child_of int,
primary key (dept_id),
foreign key (child_of) references department(dept_id) on delete
restrict on update cascade
);

create table product
( prod_id int NOT NULL,
title varchar(50) NOT NULL,
prod_desc varchar(200) NOT NULL,
short_desc varchar(50) NOT NULL,
featured char(1) NOT NULL,
price_retail_no_tax float NOT NULL,
tax_percent float NOT NULL,
discount_perc float NOT NULL,
stock int NOT NULL,
dept_id int NOT NULL,
primary key (prod_id),
foreign key (dept_id) references department(dept_id) on delete
restrict on update cascade
```

```
);

create table product_keywords
( keyword varchar(20) NOT NULL,
prod_id int NOT NULL,
primary key (keyword, prod_id),
foreign key (prod_id) references product(prod_id) on delete
restrict on update cascade
);

create table orders
( order_id int NOT NULL,
date date NOT NULL,
payment_code char(64) NOT NULL,
tracking_nbr varchar(30),
customer_ssn char(12) NOT NULL,
primary key (order_id),
foreign key (customer_ssn) references users(ssn) on delete
restrict on update cascade
);

create table order_detail
(order_id int NOT NULL,
prod_id int NOT NULL,
qty int NOT NULL,
order_price_no_tax float NOT NULL,
order_tax_percent float NOT NULL,
order_discnt_percent float NOT NULL,
primary key (order_id, prod_id),
foreign key (order_id) references orders(order_id) on delete
restrict on update cascade,
foreign key (prod_id) references product(prod_id) on delete
restrict on update cascade
);

create table reviews
( ssn char(12) NOT NULL,
prod_id int NOT NULL,
stars_nbr int NOT NULL,
long_text varchar(200) NOT NULL,
date date NOT NULL,
primary key (ssn, prod_id),
foreign key (ssn) references users(ssn) on delete restrict on
update cascade,
foreign key (prod_id) references product(prod_id) on delete
restrict on update cascade
);

create table order_status_hist
( status varchar(10) NOT NULL,
order_id int NOT NULL,
```

```
date date NOT NULL,
time time NOT NULL,
primary key (status, order_id),
foreign key (order_id) references orders(order_id) on delete
restrict on update cascade
);
```

After creating the tables, the following figure shows the schema obtained from the DBMS:



Finally, some example data is needed to use the new database. There is also a SQL statement to load data to the tables as follows:

INSERT DATA INTO TABLE “department”:

```
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(1,'root','Welcome to Uumazon!','Welcome to Uumazon',NULL);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(2,'Electronics','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum fermentum. Duis
eleifend tellus semper libero vehicula convallis. Cras
scelerisque nibh id quam placerat.','Lorem Ipsum is simply dummy
text.',1);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(3,'Clothing','Lorem ipsum dolor sit amet, consectetur adipiscing
```

```
elit. Aenean at arcu eu sem elementum fermentum. Duis eleifend
tellus semper libero vehicula convallis. Cras scelerisque nibh id
quam placerat.','Lorem Ipsum is simply dummy text.',1);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(4,'Home','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum. Duis eleifend
tellus semper libero vehicula convallis. Cras scelerisque nibh id
quam placerat.','Lorem Ipsum is simply dummy text.',1);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(5,'Computers and Tablets','Lorem ipsum dolor sit amet,
consectetur adipiscing elit. Aenean at arcu eu sem elementum
fermentum. Duis eleifend tellus semper libero vehicula convallis.
Cras scelerisque nibh id quam placerat.','Lorem Ipsum is simply
dummy text.',2);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(6,'TV and Video','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum fermentum. Duis
eleifend tellus semper libero vehicula convallis. Cras
scelerisque nibh id quam placerat.','Lorem Ipsum is simply dummy
text.',2);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(7,'Mobile Phones','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum fermentum. Duis
eleifend tellus semper libero vehicula convallis. Cras
scelerisque nibh id quam placerat.','Lorem Ipsum is simply dummy
text.',2);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(8,'Men Clothing','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum fermentum. Duis
eleifend tellus semper libero vehicula convallis. Cras
scelerisque nibh id quam placerat.','Lorem Ipsum is simply dummy
text.',3);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(9,'Women Clothing','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum fermentum. Duis
eleifend tellus semper libero vehicula convallis. Cras
scelerisque nibh id quam placerat.','Lorem Ipsum is simply dummy
text.',3);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(10,'Bedroom','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum. Duis eleifend
```

```
tellus semper libero vehicula convallis. Cras scelerisque nibh id
quam placerat.','Lorem Ipsum is simply dummy text.',4);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(11,'Kitchen','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum. Duis eleifend
tellus semper libero vehicula convallis. Cras scelerisque nibh id
quam placerat.','Lorem Ipsum is simply dummy text.',4);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(12,'Bathroom','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum fermentum. Duis
eleifend tellus semper libero vehicula convallis. Cras
scelerisque nibh id quam placerat.','Lorem Ipsum is simply dummy
text.',4);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(13,'Laptops','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum. Duis eleifend
tellus semper libero vehicula convallis. Cras scelerisque nibh id
quam placerat.','Lorem Ipsum is simply dummy text.',5);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(14,'Desktops','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum fermentum. Duis
eleifend tellus semper libero vehicula convallis. Cras
scelerisque nibh id quam placerat.','Lorem Ipsum is simply dummy
text.',5);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(15,'Monitors','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum fermentum. Duis
eleifend tellus semper libero vehicula convallis. Cras
scelerisque nibh id quam placerat.','Lorem Ipsum is simply dummy
text.',5);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(16,'Accessories','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum fermentum. Duis
eleifend tellus semper libero vehicula convallis. Cras
scelerisque nibh id quam placerat.','Lorem Ipsum is simply dummy
text.',5);
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(17,'LED TVs','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum. Duis eleifend
tellus semper libero vehicula convallis. Cras scelerisque nibh id
quam placerat.','Lorem Ipsum is simply dummy text.',6);
```

```
insert into
department(dept_id,title,dept_desc,short_desc,child_of) values
(18,'Home Theater','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum fermentum. Duis
eleifend tellus semper libero vehicula convallis. Cras
scelerisque nibh id quam placerat.','Lorem Ipsum is simply dummy
text.',6);
```

INSERT DATA INTO TABLE “product”:

```
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (1,'Lenovo
Y5','Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',14000,0.25,0.05,2,13);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (2,'Lenovo
Yoga','Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',14500,0.25,0.15,5,13);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (3,'Lenogo
Legion','Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','1',12000,0.25,0,3,14);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(4,'Samsung 21"','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum
fermentum.','Lorem Ipsum is simply dummy
text.','0',1000,0.25,0.07,10,15);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(5,'Keyboard & Mouse','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum
fermentum.','Lorem Ipsum is simply dummy
text.','0',300,0.25,0.1,25,16);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (6,'LG LED
TV 32 inch','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',1200,0.25,0.12,15,17);
insert into
```

```
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (7,'LG LED
TV 40 inch','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',1800,0.25,0.02,14,17);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (8,'LG LED
TV 50 inch','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','1',2500,0.25,0.05,18,17);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(9,'BlueRay Philips','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum
fermentum.','Lorem Ipsum is simply dummy
text.','1',700,0.25,0.05,28,18);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(10,'Samsung S25 Ultra','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum
fermentum.','Lorem Ipsum is simply dummy
text.','1',15000,0.25,0.1,34,7);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(11,'Samsung Fold 1','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum
fermentum.','Lorem Ipsum is simply dummy
text.','1',13000,0.25,0.08,31,7);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(12,'iPhone 17 Pro Max','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum
fermentum.','Lorem Ipsum is simply dummy
text.','1',15000,0.25,0.15,10,7);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(13,'iPhone 17 Pro','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum
fermentum.','Lorem Ipsum is simply dummy
text.','0',13000,0.25,0.09,15,7);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
```

```
no_tax,tax_percent,discount_perc,stock,dept_id) values
(14,'iPhone 5','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum
fermentum.','Lorem Ipsum is simply dummy
text.','0',5000,0.25,0.04,2,7);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (15,'Men
Pants','Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',200,0.25,0.08,40,8);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (16,'Men
Underwear','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',100,0.25,0.11,56,8);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (17,'Men
T-shirts','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','1',150,0.25,0.09,82,8);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (18,'Men
Shoes','Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',400,0.25,0.04,4,8);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (19,'Women
Pants','Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','1',250,0.25,0.03,45,9);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (20,'Women
Underwear','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',150,0.25,0.03,95,9);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values (21,'Women
T-shirts','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',200,0.25,0.11,64,9);
insert into
```

```
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(22,'Bed','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',900,0.25,0.02,8,10);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(23,'Mattress','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum
fermentum.','Lorem Ipsum is simply dummy
text.','1',700,0.25,0.13,8,10);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(24,'Duvet','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',400,0.25,0.07,19,10);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(25,'Pan','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',200,0.25,0.08,34,11);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(26,'Oven','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','0',1500,0.25,0.05,7,11);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(27,'Microwave Oven','Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Aenean at arcu eu sem elementum
fermentum.','Lorem Ipsum is simply dummy
text.','1',1200,0.25,0.07,8,11);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(28,'Mirror','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
simply dummy text.','1',500,0.25,0.13,4,12);
insert into
product(prod_id,title,prod_desc,short_desc,featured,price_retail_
no_tax,tax_percent,discount_perc,stock,dept_id) values
(29,'Towel','Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Aenean at arcu eu sem elementum fermentum.','Lorem Ipsum is
```

```
simply dummy text.','1',150,0.25,0.12,24,12);
```

INSERT DATA INTO TABLE “users”:

```
insert into
users(ssn,first_name,last_name,street,house_nbr,phone_nbr,news_pe
rm,password,zip_code,city,email) values
('200110202445','Stefano','Marangoni','Kungsgatan','24','75426355
4','1','stef123*','75414','Uppsala','stefano@marangoni.it');
insert into
users(ssn,first_name,last_name,street,house_nbr,phone_nbr,news_pe
rm,password,zip_code,city,email) values
('200005281224','Elias','Kassie','Flogsta','4','764521872','0','e
l11**','79844','Uppsala','elias@kassie.et');
insert into
users(ssn,first_name,last_name,street,house_nbr,phone_nbr,news_pe
rm,password,zip_code,city,email) values
('200204081221','Victor','Cornejo','Klostergatan','16','765425869
','0','vic987**','74586','Uppsala','victor@cornejo.pe');
insert into
users(ssn,first_name,last_name,street,house_nbr,phone_nbr,news_pe
rm,password,zip_code,city,email) values
('200003204112','Juan','Carrasco','Fredrika
Bremersgatan','24','767575401','1','jua321*','78945','Uppsala','j
uan@carrasco.bo');
```

INSERT DATA INTO TABLE “orders”:

```
insert into
orders(order_id,date,payment_code,tracking_nbr,customer_ssn)
values (1, DATE '2025-12-
02','ABC52355DJMRSJJJSRS1254324ERT24555TTHNSJAESS45321625EASF3222
1512','AI2542258','200005281224');
insert into
orders(order_id,date,payment_code,tracking_nbr,customer_ssn)
values (2, DATE '2025-12-
02','ABC52355DJMRSJJJSRS1254324ERT24555TTHNSJAESS45321625EASF3222
5845','AM6585548','200003204112');
```

INSERT DATA INTO TABLE “order_detail”:

```
insert into
order_detail(order_id,prod_id,qty,order_price_no_tax,order_tax_pe
rcent,order_discnt_percent) values (1,7,1,1800,0.25,0.1);
insert into
order_detail(order_id,prod_id,qty,order_price_no_tax,order_tax_pe
rcent,order_discnt_percent) values (1,15,2,200,0.25,0.5);
insert into
order_detail(order_id,prod_id,qty,order_price_no_tax,order_tax_pe
```

```
rcent,order_discnt_percent) values (1,25,1,200,0.25,0.5);
insert into
order_detail(order_id,prod_id,qty,order_price_no_tax,order_tax_pe
rcent,order_discnt_percent) values (2,16,3,100,0.25,0.2);
insert into
order_detail(order_id,prod_id,qty,order_price_no_tax,order_tax_pe
rcent,order_discnt_percent) values (2,29,2,150,0.25,0.12);
```

INSERT DATA INTO TABLE “order_status_hist”:

```
insert into order_status_hist(status,order_id,date,time) values
('new',1, DATE '2025-12-02', TIME '10:44:00');
insert into order_status_hist(status,order_id,date,time) values
('open',1, DATE '2025-12-02', TIME '11:25:00');
insert into order_status_hist(status,order_id,date,time) values
('dispatched',1, DATE '2025-12-03', TIME '13:14:00');
insert into order_status_hist(status,order_id,date,time) values
('new',2, DATE '2025-12-02', TIME '16:21:00');
insert into order_status_hist(status,order_id,date,time) values
('open',2, DATE '2025-12-03', TIME '09:57:00');
```

INSERT DATA INTO TABLE “reviews”:

```
insert into reviews(ssn,prod_id,stars_nbr,long_text,date) values
('200005281224',15,4,'very good pant, soft and confortable', DATE
'2025-12-05');
insert into reviews(ssn,prod_id,stars_nbr,long_text,date) values
('200003204112',16,5,'exact fit, fast arriving', DATE '2025-12-
05');
```

INSERT DATA INTO TABLE “product_keywords”:

```
insert into product_keywords(keyword,prod_id) values
('laptop',1);
insert into product_keywords(keyword,prod_id) values ('gamer',1);
insert into product_keywords(keyword,prod_id) values
('laptop',2);
insert into product_keywords(keyword,prod_id) values ('gamer',2);
insert into product_keywords(keyword,prod_id) values ('flat',4);
insert into product_keywords(keyword,prod_id) values
('widescreen',4);
insert into product_keywords(keyword,prod_id) values ('home',9);
insert into product_keywords(keyword,prod_id) values
('theater',9);
insert into product_keywords(keyword,prod_id) values
('apple',12);
insert into product_keywords(keyword,prod_id) values
('boxers',16);
insert into product_keywords(keyword,prod_id) values
```

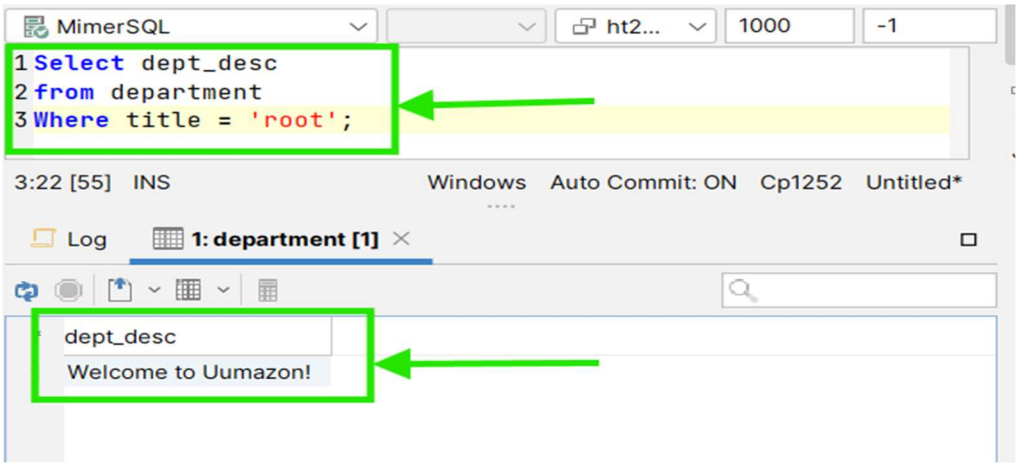
```
('feathers',24);
insert into product_keywords(keyword,prod_id) values
('fiber',24);
```

5 SQL queries

5.1 Mandatory queries

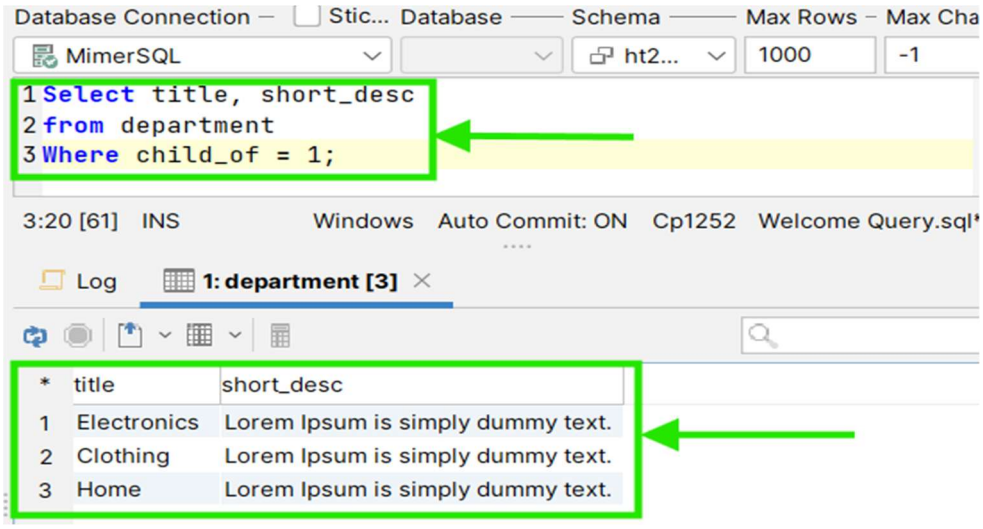
- Welcome text for the homepage:

```
Select dept_desc
from department
Where title = 'root';
```



- List of the top-level departments with fields needed for the homepage:

```
Select title, short_desc
from department
Where child_of = 1;
```



- List of the featured products with fields needed for the homepage:

```
select title, short_desc, (price_retail_no_tax * (1+tax_percent)) as
current_retail_price
from product
where featured = '1'
```

Log | 1: product [12]

*	title	short_desc	current_retail_price
1	Lenogo Legion	Lorem Ipsum is simply dummy text.	15000.0
2	LG LED TV 50 inch	Lorem Ipsum is simply dummy text.	3125.0
3	BlueRay Philips	Lorem Ipsum is simply dummy text.	875.0
4	Samsung S25 Ultra	Lorem Ipsum is simply dummy text.	18750.0
5	Samsung Fold 1	Lorem Ipsum is simply dummy text.	16250.0
6	iPhone 17 Pro Max	Lorem Ipsum is simply dummy text.	18750.0
7	Men T-shirts	Lorem Ipsum is simply dummy text.	187.5
8	Women Pants	Lorem Ipsum is simply dummy text.	312.5
9	Mattress	Lorem Ipsum is simply dummy text.	875.0
10	Microwave Oven	Lorem Ipsum is simply dummy text.	1500.0
11	Mirror	Lorem Ipsum is simply dummy text.	625.0
12	Towel	Lorem Ipsum is simply dummy text.	187.5

Here we have decided that featured products will have “featured” field filled with ‘1’, for all the other the value will be ‘0’.

- Given a product, list all keyword-related products

```
select distinct p.prod_id, title
from (product p join product_keywords pk on p.prod_id = pk.prod_id)
where pk.keyword in (
select keyword
from product_keywords
where prod_id = 2) and pk.prod_id != 2;
```

Database Connection — Sticky Database — Schema — Max Rows — Max Chars

MimerSQL | ht25_2_1dl... | 1000 | -1

```
1 select distinct p.prod_id, title
2 from (product p join product_keywords pk on p.prod_id = pk.prod_id)
3 where pk.keyword in (
4 select keyword
5 from product_keywords
6 where prod_id = 2) and pk.prod_id != 2
```

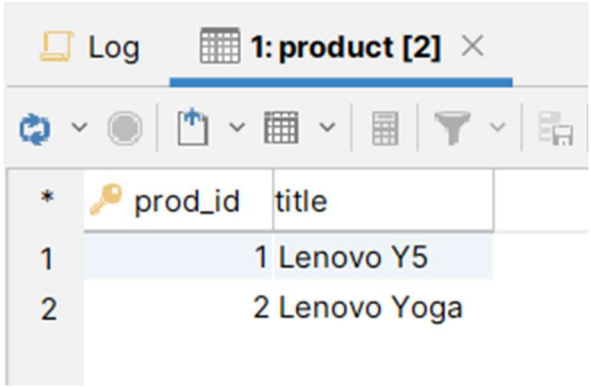
6:39 [199] INS Windows Auto Commit: ON Cp1252 Welcome Query.sql*

Log | 1: product_keywords [1]

*	prod_id	title
1	1	Lenovo Y5

- Given a product, list all keyword-related products (alternative version):

```
select prod_id, title from product where prod_id in
(
 select
  distinct prod_id
 from
  product_keywords
 where
 keyword in
 (select
  keyword
 from
  product,
  product_keywords
 where
  product.prod_id = 1 and
  product.prod_id = product_keywords.prod_id
 )
```



With the second version we just wanted to try a more "readable" query, since the first one is compact (and it might have better performance with huge amounts of data) but with the second one it's easier to follow the various steps that we need to go through to get the final result.

Moreover, in the first version we are searching for products sharing the same keywords (or some) as product 2, but we also want to cut product 2 from the resulting list.
In the alternative version we are keeping in the result also the starting product.

- Given a department, list of all its products (title, short description, current retail price) with their average rating

```
select
 department.title as department,
 product.product_id as prod_id,
 product.title as title,
 product.short_desc as short_desc,
 avg(product.price_retail_no_tax * (1+product.tax_percent)) as
current_retail_price,
 avg(reviews.stars_nbr) as average_rating
from
 department,
 product,
 reviews
where
 department.dept_id = 8 and
 product.dept_id = department.dept_id and
 reviews.prod_id = product.prod_id
group by
 department.title,
 product.product_id,
 product.title,
 product.short_desc
```

Log | 1: department [2]

*	department	title	short_desc	current_retail_price	average_rating
1	Men Clothing	Men Pants	Lorem Ipsum is simply dummy text.	250.0	4
2	Men Clothing	Men Underwear	Lorem Ipsum is simply dummy text.	125.0	5

The average star-review for each product is calculated with the instruction in the 'select' block. Also the retail price is calculated using the product.price_retail_no_tax and product.tax_percent inside the 'select' block.

- List of all products on sale sorted by the discount percentage (starting with the biggest discount)

```
select
 prod_id,
 title,
 discount_perc
from
 product
```

```
where
 discount_perc > 0
order by
 discount_perc desc
```

Log | 1: product [20]

*	prod_id	title	discount_perc
1	12	iPhone 17 Pro Max	0.15
2	23	Mattress	0.13
3	28	Mirror	0.13
4	29	Towel	0.12
5	16	Men Underwear	0.11
6	5	Keyboard & Mouse	0.1
7	10	Samsung S25 Ultra	0.1
8	17	Men T-shirts	0.09
9	11	Samsung Fold 1	0.08
10	15	Men Pants	0.08
11	4	Samsung 21"	0.07
12	24	Duvet	0.07
13	27	Microwave Oven	0.07
14	8	LG LED TV 50 inch	0.05
15	9	BlueRay Philips	0.05
16	26	Oven	0.05
17	14	iPhone 5	0.04
18	19	Women Pants	0.03
19	20	Women Underwear	0.03
20	22	Bed	0.02

5.2 Optional queries

- List 10 best-selling products (in last 30 days)

```
select
 product.prod_id as prod_id,
 product.title as title,
 sum(order_detail.qty) as qty
from
 product,
 order_detail,
 orders
where
 order_detail.prod_id = product.prod_id and
 orders.order_id = order_detail.order_id and
 orders.date >= CURRENT_DATE - INTERVAL'30'DAY
group by
 product.prod_id,
 product.title
order by
 qty desc
fetch first 10
```

Log 1: product [5]

*	prod_id	title	qty
1	16	Men Underwear	3
2	15	Men Pants	2
3	29	Towel	2
4	7	LG LED TV 40 inch	1
5	25	Pan	1

In this optional query we are selecting the products that were sold the most in the last 30 days (based on quantity, not value).
We join order_detail and product based on prod_id, and orders and order_detail based on order_id (but using only the orders that have a more recent date that current_date – 30, so orders that were created in the lat 30 days).
qty is calculated based on the prod_id, the result is ordered by qty in descending order and only the first 10 rows are kept.

6 Performance Optimization

The logic we followed to identify which queries we wanted/needed to optimize is based on the number of visits that the query performed.

We did not use the process time of the query since, with so little data, that information its not meaningful. But with a database populated with much more data, process time would have been a factor to consider.

6.1 Optimization 1

The first optimization has been done on the department_title column:

```
create index "department_title_index"
on "department" ( "title")
```

The difference in performance can be seen in the two screenshots below, showing the output of the EXPLAIN command performed on:

```
Select title, short_desc
from department
Where child_of = 1;
```

Log | 1: EXPLAIN select dept_desc from ... [2]

*	ID	PARENT	OPERATION	OPERATIONTYPE	SCANORDER	ACC_COST	HITS	VISITS	TABLE	ALIAS	INDEX	INDEXONLY
1	1	0	select	(null)	(null)	16	1	16	(null)	(null)	(null)	(null)
2	2	1	full table scan	sequential	1	16	1	16	department	(null)	SQL_PRIMARY_KEY_0000039246	(null)

Log | 1: EXPLAIN select dept_desc from ... [2]

*	ID	PARENT	OPERATION	OPERATIONTYPE	SCANORDER	ACC_COST	HITS	VISITS	TABLE	ALIAS	INDEX	INDEXONLY
1	1	0	select	(null)	(null)	2	1	2	(null)	(null)	(null)	(null)
2	2	1	index scan, table lookup	leading keys	1	2	1	2	department	(null)	department_title_index	false

6.2 Optimization 2

The second optimization has been done on the product.featured column:

create index "product_featured_index"
on "product" ("featured")

The difference in performance can be seen in the two screenshots below, showing the output of the EXPLAIN command performed on::

```
select title, short_desc, (price_retail_no_tax * (1+tax_percent)) as
current_retail_price
from product
where featured = '1'
```

Log | 1: EXPLAIN select title, short_de... [2]

*	ID	PARENT	OPERATION	OPERATIONTYPE	SCANORDER	ACC_COST	HITS	VISITS	TABLE	ALIAS	INDEX	INDEXONLY
1	1	0	select	(null)	(null)	30	1	30	(null)	(null)	(null)	(null)
2	2	1	full table scan	sequential	1	30	1	30	product	(null)	SQL_PRIMARY_KEY_0000039259	(null)

Log | 1: EXPLAIN select title, short_de... [2]

*	ID	PARENT	OPERATION	OPERATIONTYPE	SCANORDER	ACC_COST	HITS	VISITS	TABLE	ALIAS	INDEX	INDEXONLY
1	1	0	select	(null)	(null)	2	1	2	(null)	(null)	(null)	(null)
2	2	1	index scan, table lookup	leading keys	1	2	1	2	product	(null)	product_featured_index	false

6.3 Optimization 3

The third optimization has been done on the product.discount_perc column:

create index "product_discount_index"
on "product" ("discount_perc")

The difference in performance can be seen in the two screenshots below, showing the output of the EXPLAIN command performed on:

```
select
 prod_id,
 title,
 discount_perc
from
 product
where
 discount_perc > 0
order by
 discount_perc desc
```

Log | 1: EXPLAIN select prod_id, title,... [3]

*	ID	PARENT	OPERATION	OPERATIONTYPE	SCANORDER	ACC_COST	HITS	VISITS	TABLE	ALIAS	INDEX	INDEXONLY
1	1		0 select	(null)	(null)	54	10	40	(null)	(null)	(null)	(null)
2	2		1 sorting	temporary table order by	(null)	(null)	(null)	10	(null)	(null)	(null)	(null)
3	3		2 full table scan	sequential	1	30	10	30	product	(null)	SQL_PRIMARY_KEY_0000039259	(null)

Log | 1: EXPLAIN select prod_id, title,... [2]

*	ID	PARENT	OPERATION	OPERATIONTYPE	SCANORDER	ACC_COST	HITS	VISITS	TABLE	ALIAS	INDEX	INDEXONLY
1	1		0 select	(null)	(null)	20	10	20	(null)	(null)	(null)	(null)
2	2		1 index scan, table lookup	leading keys	1	20	10	20	product	(null)	product_discount_index	false

6.4 BTree or Hash indexing?

Here we are talking about two very different approaches to indexing. In the case of BTrees, we find a structure composed of nodes in which the leaves are pointers to the columns of a table.

The advantage is that it is a balanced tree, which can be searched using the classic logic belonging to binary trees, it is much faster to search for an entry, since the search time goes from O(n) to O(log(n)). For example, when searching for a name in the name column of a table with 1B rows, we will read a maximum of about 20 rows.

The disadvantage is that the index of a table takes up practically the same space as the table, sometimes even more, increasing the size of the DB. In addition, insert/update/delete operations are also slowed down, since changes must be made to both the actual table and the index.

On the other hand, a hash index uses a hash function to encode the value contained in the column. In this way, when we pass the value we are looking for to the function, we obtain the exact position of our target within the index. This means that in most cases, the search is in O(1).

The major disadvantage of the hash index is that no particular order of the columns is maintained, and it can only be used for database queries of the type “WHERE column = xxx”, but operations such as >, <, LIKE, BETWEEN are not supported.

Comparing the two methods, Btree indexes are more suitable in several cases, where it is important/interesting to order the values and where interval and similarity searches are expected (<,>, LIKE, BETWEEN). However, it is essential to choose carefully how many and which columns to index, so as not to risk exploding the size of the DB.

On the other hand, hash indexes are very useful when you are only interested in specific searches. It should also be added that in order to maintain the main advantage, i.e. O(1) searching, the values passed to the hash function must have good variability to avoid collisions as much as possible. This is because if many similar values are assigned to the same hash bucket, we will be forced to scroll through a list to find the right one, thus slowing down DB operations.

7 Python Programs

Below the two python programs to complete task 7.

The two full python files are updated in Studium. Another file named “Instructions” will contain the instructions on how to run the programs.

7.1 Department_browser.py

After entering their username and password, the user can enter a department ID.

```
PS C:\Users\Stefano\DB1> python department_browser.py
Enter your Studium username: stma2590
Enter your Studium password A:

Enter department ID to show child depts or products:
```

is_leaf is the method that checks whether the dept_id appears in the child_of column. If it does not appear, the department is a leaf and the result is true

- If the department is a leaf, the *get_products* method returns a list (in table form) of all the products contained in that department.

```
Enter department ID to show child depts or products: 8


Leaf department, listing products:

   ID          TITLE  PRICE WO/TAX  TAX %  DISCNT %  RETAIL PRICE
0  15      Men Pants         200.0   0.25      0.08        230.00
1  16  Men Underwear         100.0   0.25      0.11        111.25
2  17   Men T-shirts         150.0   0.25      0.09        170.63
3  18      Men Shoes         400.0   0.25      0.00        500.00

Do you want to exit?  (Y/N):
```

- If the department is not a leaf, the *get_child_depts* method is called and the programme returns a list of all child departments.

```
Enter department ID to show child depts or products: 6


Not leaf department, listing child departments:

   ID         TITLE
0  17       LED TVs
1  18  Home Theater

Do you want to exit?  (Y/N):
```

After the result is shown, the user can decide to keep examining other departments or to exit the program.
To exit the program, digit ‘y’ or ‘Y’ when the question “*Do you want to exit? (Y/N):*” is shown.

7.2 Product_update.py

In this case, we decided to expand the programme's functionality described in the task.
After entering a product code, the user is presented with a list of available updates (not just the discount% update).

```
PS C:\Users\Stefano\DB1> python product_update.py
Enter your Studium username: stma2590
Enter your Studium password A:

Enter product ID to show details: 12

=====================================================
Product ID  : 12
Title       : iPhone 17 Pro Max
Price wo/tax: 15000.0
Tax percent : 25.0%
Discount    : 30.0%
Retail Price: 13125.0
Featured    : 1
=====================================================


What do you want to do?
1. Update title
2. Update price before tax
3. Update tax percent
4. Update discount percent
5. Update featured
9. Select another product
0. Exit program

Enter the option number:
```

User will digit:

- 1 → to update the title
- 2 → to update the price before tax
- 3 → to update the tax%
- 4 → to update the discount%
- 5 → to update the is_featured status
- 9 → to be able to select another product if no update is needed on current product
- 0 → to exit the program if no action in needed

```
Enter the option number: 3
Enter new tax percent (0-100): 23
Updated! 1 row(s) affected.
New information:

=====================================================
Product ID  : 12
Title       : iPhone 17 Pro Max
Price wo/tax: 15000.0
Tax percent : 23.0%
Discount    : 30.0%
Retail Price: 12915.0
Featured    : 1
=====================================================

Do you want to exit?  (Y/N): n
```

After each update operation the user is asked if they want to terminate the program. By typing 'Y' or 'y', the program is terminated.

get_product is the method that retrieves product information from the database. This information is then showed to the user that can decide if something needs to be updated.

*update_* methods are used to manage each different update type.

7.3 Error and Exception handling

Errors and Exceptions are managed inside try/except blocks using the standard Error management available in mimerpy.

If the user tries to select a dept_id or prod_id that is not available inside the database, mimerpy does not raise any Exception, instead it just returns an empty result (len(curson.fetchall()) == 0).
This case is managed in the code by “manually” raising an Exception.

```
Enter product ID to show details: 77

=========================================================


Exception occurred: list index out of range

Do you want to exit?  (Y/N): n
```